

DNN Closed-Book 100 Marks Exam Bible

Exam Strategy

Spend 60% time on repeated PYQs. Master formulas and derivations. Solve numericals without notes.

Golden Formulas

$z = Wx + b$; $\text{sigmoid} = 1/(1 + e^{-z})$; $dL/dz = \hat{y} - y$; $\text{gradient} = (\hat{y} - y)x$; $\text{softmax} = \exp(z_i) / \sum(\exp(z_j))$.

Perceptron

Algorithm: compute z , apply step function, compare with label, update $w_{\text{new}} = w + \eta(y - \hat{y})x$.

Logistic Regression Worked Example

Given $z = 1.2$, $\text{sigmoid} = 0.768$. If $y = 1$ then $\text{error} = -0.232$ and $\text{gradient} = (\hat{y} - y)x$.

BCE Derivation

$\text{Loss} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \Rightarrow dL/dz = \hat{y} - y$.

Softmax+CCE

Compute exponentials, normalize, $\text{loss} = -\log(\text{probability of true class})$, $\text{gradient} = \hat{y} - y$.

Forward Propagation

$Z1 = W1X + b1$; $A1 = g(Z1)$; $Z2 = W2A1 + b2$; $A2 = \text{sigmoid}(Z2)$.

Backpropagation

$dZ2 = A2 - Y$; $dW2 = dZ2 * A1^T$; $dZ1 = (W2^T dZ2) * g'(Z1)$.

CNN Output

$\text{Output} = (N - F + 2P) / S + 1$. Example 8×8 with 3×3 gives 6×6 .

CNN vs ANN

CNN uses local receptive fields and parameter sharing.

Transfer Learning

Use pretrained model for small datasets to reduce overfitting.

RNN

$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t)$.

GRU

$h_t = z_t h_{t-1} + (1 - z_t) \tilde{h}_t$.

Attention

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$.

Transformer

Encoder for understanding, decoder for generation, encoder-decoder for summarization.

Optimizers

SGD slow; Momentum faster; RMSProp adaptive; Adam combines momentum and RMSProp.

Regularization

Dropout, L2, early stopping, batch norm.

Weak Areas

Jacobian maps vector derivatives. Chain rule propagates gradients backward.

Exam Templates

For derivations write: define loss, differentiate activation, apply chain rule, simplify.

Practice Question 1

Solve manually: derive gradients, compute outputs, update weights, explain concepts in 5 points.

Practice Question 2

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Practice Question 3

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Practice Question 4

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Practice Question 5

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Practice Question 6

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Practice Question 7

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Practice Question 8

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Practice Question 10

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Practice Question 14

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Practice Question 15

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Practice Question 16

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Practice Question 18

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Practice Question 19

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Practice Question 20

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